

A PROTOTYPE OF RECONFIGURABLE INTELLIGENT SURFACE WITH CONTINUOUS CONTROL OF THE REFLECTION PHASE

Roman Fara, Philippe Ratajczak, Dinh-Thuy Phan-Huy, Abdelwaheb Ourir, Marco Di Renzo, and Julien de Rosny

ABSTRACT

With the development of the next generation of mobile networks, new research challenges have emerged, and new technologies have been proposed to face them. On one hand, the reconfigurable intelligent surface (RIS) technology is being investigated for partially controlling the wireless channels. RIS is a promising technology for improving the signal quality by controlling the propagation of the electromagnetic waves in a nearly passive manner. On the other hand, ambient backscatter communications (AmBC) is another promising technology that is tailored to address the energy efficiency requirements of the Internet of Things. AmBC enables low-power communications by backscattering the ambient signals and thus reusing existing electromagnetic waves for communications. RIS can be utilized in the context of AmBC for improving system performance. In this article, we report a prototype of an RIS that offers the capability of controlling the phase shifts applied to the incident waves in a continuous manner, and we characterize its properties with the aid of full-wave simulations and through experimental measurements. Specifically, we introduce a phase shift model for predicting the signal reflected by the designed RIS prototype. We apply the proposed model for optimizing an RIS-assisted AmBC system, and we demonstrate that the use of an RIS can significantly improve system performance.

INTRODUCTION

Each generation of mobile networks has improved the efficiency of wireless communications. From the second generation (2G) to fifth generation (5G) mobile networks, the performance of wireless communication systems has improved, thanks to the research and development efforts on the design and optimization of transmitters, receivers, and transmission protocols. However, the propagation channel separating the transmitters and receivers has inherently been perceived as not controllable and sometimes unfavorable to the propagation of the electromagnetic (EM) waves. This is one of the challenges that have been identified for next generation mobile networks [1].

Recently, the reconfigurable intelligent surface (RIS) technology has been proposed and investigat-

ed for application to wireless communications [2]. An RIS is a planar structure that comprises a large number of unit cells that are capable of scattering (e.g., reflecting and/or refracting) the EM waves in a controlled manner [3]. An RIS can modify the phase, the amplitude, or even the polarization of the incident EM waves thanks to a smart controller that adjusts the response of each unit cell. Notably, an RIS is nearly passive, as it reradiates the EM waves without requiring power amplifiers and digital signal processing units. Some energy is only required for the smart controller and to enable the reconfigurability of the RIS. Therefore, an RIS can improve signal reliability or mitigate interference in a nearly passive manner, without additional densification of multiple-antenna transmitters and network infrastructure elements that require power amplification and signal regeneration. Since an RIS can partially control and shape the wireless channels as one desires, it shows promising potential applications in future wireless networks [3].

In particular, an RIS can be employed for enhancing the performance of another promising technology, especially in the context of the Internet of Things (IoT): ambient backscatter communications (AmBC) [4]. In AmBC, a device, named a tag, transmits data to another device, named a reader, without generating additional EM waves. In simple terms, the tag is illuminated by an ambient signal source, such as a TV tower, a WiFi hotspot, or even a base station, and this received signal is then backscattered by the tag. The tag is able to modulate the backscattered signal by implicitly transmitting its own message. The modulated signal is detected by the reader, which can decode the message of the tag. A simple tag modulates the backscattered signal according to two possible states: a backscattering state, when the tag antenna is short-circuited, and a transparent state, when the tag antenna is open-circuited [5]. Since AmBC systems inherently reuse the ambient signals and do not transmit additional EM waves for communication, they are considered a suitable technology for very low-power applications [6]. However, the tag-to-reader link may be inherently interfered by the ambient signal. Also, the tag may not backscatter a strong enough signal to the reader if, for example, it is located in a deep fade of the

ambient signal [7]. AmBC is therefore a promising technology, but it has some limitations.

To overcome these limitations, AmBC can be assisted by an RIS. For example, the authors of [8] showed that the phase shifts of the unit cells of an RIS can be optimized to compensate for the multipath effect of the propagation channel in order to reduce the interference and to increase the signal quality at the reader. In [9], the RIS plays the role of the tag and is used to backscatter the message to the reader. Since many unit cells are available at the RIS, the backscattered signal is enhanced and can be detected more easily. This principle was demonstrated experimentally in [10], and was shown to significantly improve system performance. Also, the authors of [11] report that in cognitive networks, an RIS can improve the performance of the tag-to-reader link by maximizing the signal at the location of the tag [12].

Therefore, recent research works have demonstrated that RIS is a suitable candidate technology for improving the performance of AmBC. In [12], in particular, we experimentally demonstrated that the tag-to-reader link can be enhanced by creating a passive reflected beam that improves the signal quality at the locations of the tag and reader. Similar to multiple-input multiple-output (MIMO) systems [13] but in a nearly passive manner, an RIS beamforms the reflected signal to assist the transmission of the tag. The experiments conducted in [12] were based on an RIS prototype that comprises 196 reconfigurable unit cells, each of them capable of phase shifting the reflecting signal in a controlled manner. This is realized by using varactor diodes that control the phase response of the unit cells, which ensures continuous tuning of the applied phase shifts. Even though the principle was experimentally validated in [12], the RIS was not completely characterized, and in particular, no equivalent model for the phase shift and amplitude response of the reflection coefficient was given. This is, however, essential for utilizing the RIS prototype in wireless communications.

In this article, we focus on this specific and important aspect, and introduce a simplified model for the phase and amplitude response of the reflection coefficient of a voltage-controlled RIS prototype. Based on the proposed model, we evaluate its suitability and accuracy in the context of RIS-assisted AmBC. In particular, the validation of the model is performed through full-wave simulations in Ansys HFSS and through experimental measurements. We demonstrate that the complex EM behavior of each unit cell can be modeled with simple equations that depend on the control voltage applied to the unit cells. Based on the proposed model for the unit cell, we propose a method to create a large codebook to realize passive beamforming. For given locations of the source, tag, and reader, the unit cells of the RIS are appropriately optimized to reflect and maximize the signal toward desired spots. We demonstrate that the signal scattered by the RIS can be well approximated by the linear combination of the contribution of each unit cell (based on the proposed model). Also, we prove that an RIS with a continuous phase shift tuning capability can improve the performance of AmBC. We discuss the accuracy of the proposed (simple) model for the phase shift and compare it, with the aid of full-wave simulations, against more complex models that account for fin-

er-grained modeling of the response of the unit cells. We show that the proposed simplified model is sufficiently accurate for first-order analysis of RIS-assisted communications within a given range for the angles of incidence and reflection.

The article is organized as follows. First, we introduce the RIS prototype with continuous phase shifting capability and discuss its characteristics. Then we describe a method to define a large codebook for realizing passive beamforming through controllable phase shifts. Subsequently, we illustrate the RIS-assisted AmBC principle and compare the predictions from the proposed phase shift model against accurate full-wave simulations with Ansys HFSS. Finally, we conclude the article.

RIS PROTOTYPE WITH CONTINUOUS PHASE-SHIFTING

As illustrated in Fig. 1, the RIS prototype comprises 14×14 unit cells. Each unit cell is a reflecting conductive patch that can be reconfigured independent of the others. In particular, the unit cell is a three-dimensional patch on a dielectric substrate. The patch is made of two conductive areas separated by an annular slot. The inner and outer areas are connected to each other with four MA46H120 varactor diodes. The specific arrangement of the varactor diodes ensures the adjustment of the capacitance of the unit cell, which provides the desired tunability of the reflection coefficient of the unit cell. In detail, in the reverse configuration, the capacitance of a varactor diode is inversely proportional to the junction voltage. Thus, the capacitance of the unit cell can be tuned by controlling the voltage applied to the four varactors. As illustrated in Fig. 1, the unit cell can be reconfigured by applying different voltages to the four varactor diodes, which in turn enables the configuration of the reflection coefficient.

In the realized unit cell structure, four varactors are used for each unit cell in order to ensure good symmetry of the radiation pattern. All the unit cells have identical design and size (i.e., 14×14 mm²). They are engineered to operate in the frequency range 5.15–5.75 GHz [14, 15], which yields a cell size of approximately a quarter of the signal wavelength. The unit cell structure is characterized through full-wave simulations in Ansys HFSS under locally periodic boundary conditions. In detail, the results are obtained by considering an infinitely large surface that is a periodic repetition of a unit cell configured to realize a given phase shift. Thus, a homogeneous infinite surface is simulated with the desired unit cell configuration, which ensures the correct definition of reflection coefficient.

As illustrated in Fig. 1, the desired reflection coefficient of each unit cell can be obtained through the voltage-controlled capacitance that characterizes the unit cell. More precisely, if the m th unit cell among M unit cells is illuminated by an electric field E_m^i whose direction of incidence is $\vec{u}_i$, the reflected electric field toward the direction of departure $\vec{u}_d$ can be written as $E_m^d = E_m^i \cdot r(\vec{u}_i, \vec{u}_d, v)$, where v is the voltage applied to the unit cell, and $\vec{u}_i$ and $\vec{u}_d$ are unit-norm vectors defined by the azimuth and elevation angles of incidence (θ_i, φ_i) and departure (θ_d, φ_d) , respectively, in spherical coordinates, and $r(\cdot)$ is the reflection coefficient.

To characterize the unit cell and to introduce an EM-consistent model for $r(\vec{u}_i, \vec{u}_d, v)$ that can be utilized in wireless communications, we study the

Since AmBC systems inherently reuse the ambient signals and do not transmit additional EM waves for communication, they are considered a suitable technology for very low-power applications [6]. However, the tag-to-reader link may be inherently interfered by the ambient signal.

As far as the full-wave simulations are concerned, the reflection coefficient of each unit cell is indirectly characterized as a function of the capacitance c of the varactor diodes. As far as the experimental characterization through measurements is concerned, the reflection coefficient of each unit cell is directly characterized as a function of the voltage v applied to the varactor diodes.

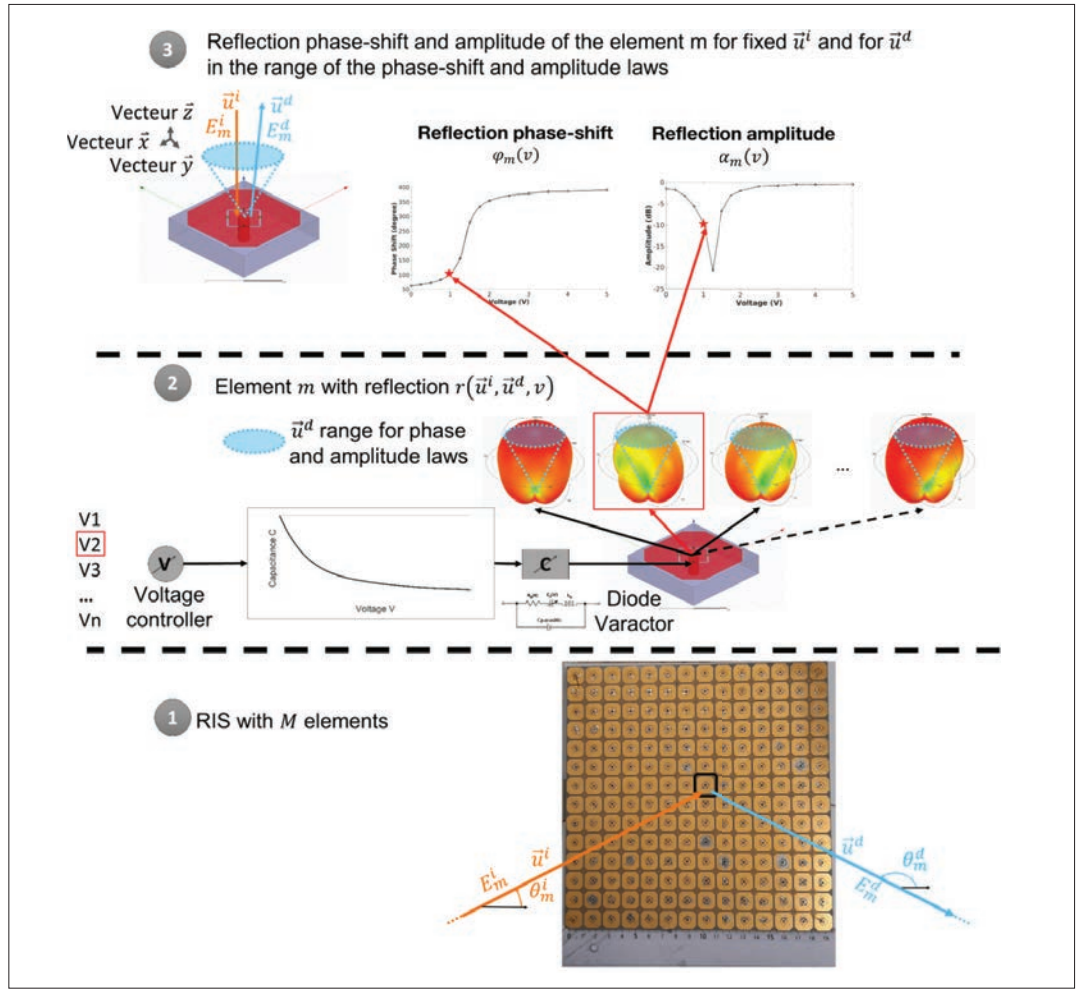


FIGURE 1. Unit cell with a voltage-controlled tunable reflection coefficient $r(\vec{u}_i, \vec{u}_d, v)$; full-wave simulation results under periodic boundary conditions [14, 15].

scattering properties of the unit cell as a function of $\vec{u}_i$, $\vec{u}_d$, and v by using the Ansys HFSS EM software under locally periodic boundary conditions and through measurements conducted with the RIS prototype. As far as the full-wave simulations are concerned, the reflection coefficient of each unit cell is indirectly characterized as a function of the capacitance c of the varactor diodes. As far as the experimental characterization through measurements is concerned, the reflection coefficient of each unit cell is directly characterized as a function of the voltage v applied to the varactor diodes. Since the capacitance of each varactor diode is voltage-controlled, as illustrated in Fig. 1, c and v are directly related to each other; that is, $v = v(c)$ or $c = c(v)$. This relation is not necessary in this study, but it can be derived through measurements.

The characterization of the reflection characteristics of the unit cell is illustrated in Fig. 2 for various elevation angles of the incident electric field. The figure reports, in particular, the phase $\varphi_m(\cdot)$ and the amplitude $\alpha_m(\cdot)$ of the reflection coefficient as a function of the capacitance of the varactor diodes (HFSS simulations) and the junction voltage (prototype experiments). From the results illustrated in Fig. 2, we evince that both the phase and the amplitude of the reflection coefficient of each unit cell are almost independent of the angle of incidence if this angle is less than 40° . Within

this range of angles, notably, the radiation pattern of the unit cell can be assumed to be almost omnidirectional. For wider angles of incidence or reflection, on the other hand, Fig. 1 shows that the radiation pattern of the unit cell is not omnidirectional, but it depends on the angle of incidence and reflection. As expected, Fig. 2 shows that the reflection coefficient can be adjusted by appropriately controlling the junction voltage in the range 0–5 V, which corresponds to a capacitance in the range 0.2–1 pF.

Therefore, by considering angles of incidence and reflection no larger than 40° with respect to the normal to the unit cell, a simple model for the reflection coefficient of the unit cells can be formulated as:

$$r(\vec{u}_i, \vec{u}_d, v) \approx r(v) \approx g_0 \alpha_m(v) e^{j\varphi_m(v)} \quad (1)$$

where g_0 denotes the unit cell (constant) gain, and $\alpha(v)$ and $\varphi(v)$ denote the amplitude and the phase of the reflection coefficient of the unit cell, respectively, which can be configured in a continuous manner as a function of the voltage v that is applied to the varactor diodes.

The approximation in Eq. 1 has been experimentally characterized with the aid of the RIS prototype, and some results based on the measurements are reported in Fig. 2. The phase and the amplitude

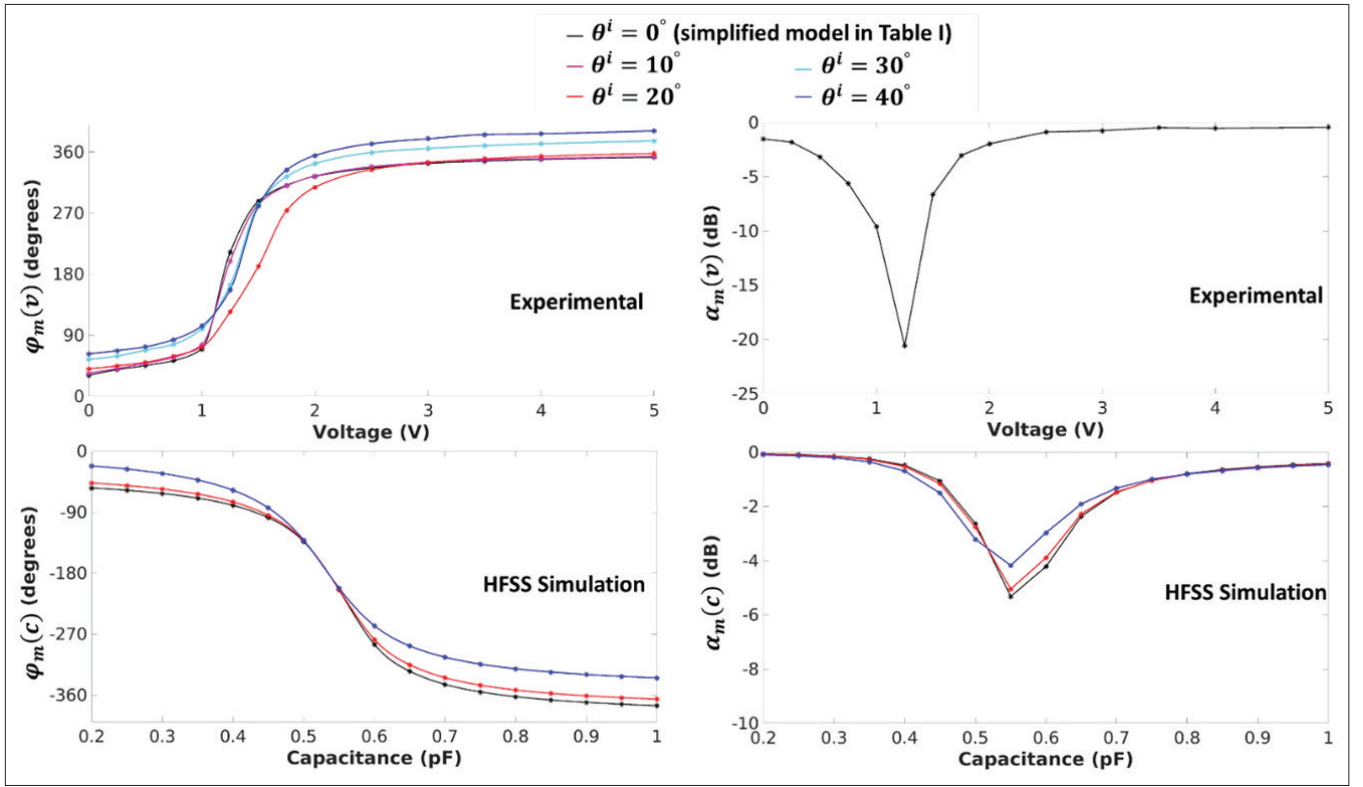


FIGURE 2. Phase and amplitude of the reflection coefficient $r(\vec{u}_i, \vec{u}_d, v)$ as a function of v for the experimental measurements and $r(\vec{u}_i, \vec{u}_d, c)$ as a function of c for HFSS simulations under periodic boundary conditions.

of the reflection coefficient depend on the angle of incidence, as illustrated in Fig. 2. If the angle of incidence is less than 40° , however, its impact is minor and can be ignored as a first-order approximation. In the rest of this article, for simplicity, we consider the reflection coefficient that corresponds to an angle of incidence equal to 0° . The corresponding amplitude $\alpha(v)$ and phase $\varphi(v)$ are reported in Table 1 as a function of the voltage v . Therefore, Table 1 is valid only for angles of incidence less than 40° for which the proposed RIS prototype was designed, optimized, and engineered. By analyzing Table 1, we observe that the amplitude and the phase of the reflection coefficient are not independent of each other, since they both depend on the control voltage. In particular, the amplitude of the reflection coefficient has the smallest value in correspondence of the range of major variation of the phase of the reflection coefficient (Fig. 2). Finally, we note that the reflection coefficient in Fig. 2 depends only on the angle of incidence (not on the angle of reflection) because of the assumption of locally periodic boundary conditions that is made to characterize it: Each value of the reflection coefficient corresponds to a homogeneous surface whose unit cells are identically configured. Therefore, only specular reflection is allowed when the reflection coefficient is characterized, that is, the angle of incidence and reflection coincide. By adjusting the phase shift of the different unit cells, however, anomalous reflection can be obtained.

In this section, we have characterized the phase and the amplitude response of the RIS prototype, and have introduced a simple model for the reflection coefficient of each unit cell, based on full-wave simulations, under the modeling assumption of locally periodic boundary conditions, and experi-

Voltage (V)	$\alpha(v)$ (dB)	$\varphi(v)$ (degrees)
0	-1.517	32.798
0.25	-1.807	40.854
0.5	-3.156	46.807
0.75	-5.59	53.543
1	-9.576	70.32
1.25	-20.563	-167.158
1.5	-6.615	-73.171
1.75	-3.029	-49.627
2	-1.959	-35.908
2.5	-0.874	-23.263
3	-0.749	-16.087
3.5	-0.469	-12.663
4	-0.528	-9.925
5	-0.439	-6.906

TABLE 1. Experimental characterization of the reflection coefficient of an elementary cell in Eq. 1: $\alpha(v)$ and $\varphi(v)$ as a function of v .

ments. We have demonstrated the main capabilities of the designed RIS prototype: the phase shift induced by each unit cell can be continuously controlled as a function of the voltage applied to the unit cell. This feature can be exploited in order to control the angle of reflection of the incidence signal and to beamform the reflected signal towards the desired direction. This is analyzed and experimentally validated next.

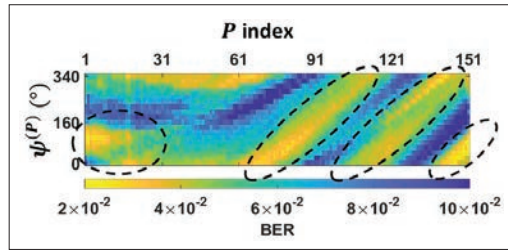


FIGURE 3. Experimental measurements of the BER for all possible values of the predefined locations ($p = 1, 2, \dots, 151$) and phase shifts $\psi^{(P)}$ (in the range 0° – 360°). The experimental setup is illustrated in Fig. 4.

RIS CODEBOOK DESIGN FOR ACCURATE BEAMSTEERING

From the EM characterization of the unit cell in Figs. 1 and 2, we evince that it is possible to treat an RIS as a blackbox that comprises several identical and voltage-controlled unit cells, each of which is capable of independently changing the amplitude and the phase of the incident EM wave. By appropriately optimizing all the unit cells of the RIS, the incident signal can be partially controlled and steered toward the desired locations in order to enhance or null the received signal in a nearly passive manner, without power amplification, without decoding and reencoding the incident EM wave, and without requiring radio frequency chains. If the operation of the RIS prototype is restricted to angles of incidence and reflection less than 40° (for which our prototype was designed), Table 1 constitutes the proxy between the EM world and the wireless communications world.

An appropriately optimized RIS can therefore steer the reflected signal toward desired directions or specified locations. Assuming, for example, that the locations of the transmitter and RIS are known, we can pre-design a codebook of beamforming vectors that provides the exact configuration of the phase shifts of every unit cell of the RIS in order to steer the reflected signal toward a receiver that is located in a given position. This is obtained by ensuring that the signals reradiated by each unit cell add up coherently in correspondence to the desired receiver location. Then, given an area of interest, a location-dependent codebook for nearly passive beamforming can be engineered.

A simple approach to realize an RIS that focuses the reflected signal toward a specified location, in line of sight of both the RIS and the transmitter, is to ensure that the phase shift applied by each unit cell is optimized to guarantee that the signals reflected by all the unit cells add up coherently at the location of the receiver, and that each unit cell adds an additional phase shift that ensures that the signal reflected by the RIS adds up coherently with the signal emitted by the transmitter at the location of the receiver. Our proposed RIS prototype enables these two functionalities simultaneously. Based on Table 1, this can be realized by controlling the junction voltage of the unit cells so as to apply a unit-cell-specific phase shift and an additional constant RIS phase shift.

More precisely, let E^S be the transmitted electric field at the transmitter (source) location S and let E^P be the RIS-reflected electric field at the predefined location P . Based on Eq. 1, under the assumption of free-space propagation, we have

$$E^P = E^S \sum_{m=1}^M \frac{\lambda e^{j \left(2\pi \frac{d_m^{S-RIS}}{\lambda} \right)}}{4\pi d_m^{S-RIS}} [g_0 \alpha_m(v) e^{j\varphi_m(v)}] \frac{\lambda e^{j \left(2\pi \frac{d_m^{RIS-P}}{\lambda} \right)}}{4\pi d_m^{RIS-P}} \quad (2)$$

where $d_m^{(S-RIS)}$ and $d_m^{(RIS-P)}$ are the transmission distances between the source and the m th unit cell of the RIS, and between the m th unit cell of the RIS and the predetermined target location P , respectively, and λ is the signal wavelength. Also, j denotes the imaginary unit, and $M = 14 \times 14 = 196$ for the considered prototype.

From Eq. 2, we evince that the phase shift that corresponds to the optical path-length associated with the m th unit cell at the target location P is as follows:

$$b_m^{(P)} = 2\pi \frac{d_m^{S-RIS} + d_m^{RIS-P}}{\lambda} \quad (3)$$

From Eq. 3, in order to ensure that the signals scattered by all the unit cells are phase-aligned with the transmitted signal at the target location P , the phase shift that each unit cell of the RIS needs to apply to the incidence signal is $\varphi_m = -b_m^{(P)} + \psi^{(P)}$, where $\psi^{(P)}$ is the phase of the transmitted signal observed at the target location P .

An illustration of the bit error rate (BER) of an RIS-assisted communication system, deployed using our RIS prototype, is illustrated in Fig. 3 as a function of several target locations (P index) and RIS phase shifts $\psi^{(P)}$. In this example, it is important to note that the 196 unit cells are clustered in 14 groups (macrocells), each of which contains 14 unit cells. More precisely, the 14 macrocells coincide with the 14 columns of the RIS prototype. This enables us to reduce the complexity of controlling the varactors of the unit cells, since the 14 unit cells of each macrocell are configured to the same phase shift, that is, the control voltage applied to them is identical. This does not significantly affect the capability of controlling the channel for the considered network topology.

A photo of the considered system setup for application to AmBC is given in Fig. 4. The experimental results in Fig. 3 are obtained by pre-designing the beamforming codebook in Eq. 3 in correspondence to 151 possible locations of a target receiver, and by then measuring the BER at the specified location for every pair of phase shifts $(b_m^{(P)}, \psi^{(P)})$ of the RIS codebook. From Fig. 3, we observe that there are multiple pairs $(b_m^{(P)}, \psi^{(P)})$ which provide good BER performance. In particular, the pairs depicted in yellow correspond to possible configurations of the RIS that enhance the link performance, as compared to the same system setup in the absence of RIS.

It is worth mentioning that the performance enhancement offered by RIS-assisted communications can be further improved (with respect to the results illustrated in Fig. 3). In fact, the beam-

forming design in Eq. 3 is sub-optimal and the simplest one that can be applied, as it accounts only for the different optical path-lengths of the signals scattered by the unit cells. Equation 2 and the considered case study are referred to a single tag scenario. Networks with multiple tags can be considered as well, in which the unit cells of the RIS are configured to create multiple beams towards the intended tags.

RIS-ASSISTED AmBC: UNIT CELL MODELING ACCURACY

In AmBC, the system performance highly depends on the power level of the signal received at the tag. Since a tag is a passive device, the amount of power that is backscattered is ultimately determined by the amount of power that the tag receives from the transmitter (the source in Fig. 4). In this context, an RIS can be utilized to passively beamform the signal emitted by the source toward the tag in order to enhance the amount of power received and thus the amount of power that is backscattered toward the receiver (the reader in Fig. 4). To this end, the RIS needs to be appropriately optimized in terms of beamforming vector and RIS phase shift; that is, the optimal pair $(b_m^{(P)}, \psi^{(P)})$ needs to be identified, where P denotes the location of the tag in the considered example.

Therefore, it is important to utilize an accurate model for the RIS in order to enable a simple but accurate design of the configuration of the unit cells. In Table 1, we have introduced a simple model for the amplitude and phase of the reflection coefficient of each individual unit cell of the RIS as a function of the control voltage. The objective of this section is to evaluate the accuracy of the model introduced in Table 1 as compared to full-wave simulations performed with Ansys HFSS.

The results of this analysis are illustrated in Fig. 5 and are obtained as follows.

Simplified Model in Table 1: Given the locations of the source, tag, RIS, and reader in Fig. 4, three pairs $(b_m^{(P)}, \psi^{(P)})$ that minimize the BER in Fig. 3 are considered. The optimal beamforming vector and RIS phase shift are identified by using the simplified model of the RIS in Table 1 and the simple analytical model for the scattered field in Eq. 2. In particular, we assume $g_0 = 1$, but we account for the fact that the amplitude of the reflection coefficient of each unit cell is not independent of the phase shift (and the unit cell control voltage).

Full-Wave Simulations in Ansys HFSS: This software enables accurate full-wave simulations based on a finite element method solver for electromagnetic structures. We simulate the entire RIS structure in Ansys HFSS by considering the actual scattering from each unit cell as illustrated in Fig. 1. The same three pairs $(b_m^{(P)}, \psi^{(P)})$ as for the simplified model are considered, but no assumption on the modeling of the unit cells is made. In particular, the analytical models in Table 1 (no periodic boundary conditions) and Eq. 2 are not assumed, and the mutual coupling among the unit cells of the RIS is accounted for. In addition, the complex structure of the unit cells, including the varactors, is accurately modeled [14, 15], the finite size of the RIS is taken into account in the full-wave simulations, and

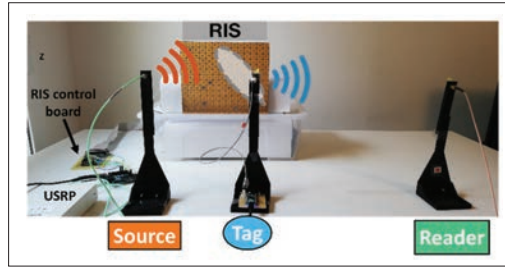


FIGURE 4. Illustration of the considered testbed for RIS-assisted AmBC.

no periodic boundary conditions at the unit cell level are applied in this case.

Figure 5 illustrates the field scattered by the RIS for three different beamforming pairs $(b_m^{(P)}, \psi^{(P)})$ that minimize the BER. The results illustrated in Fig. 5 show, as expected, that some differences exist between the predictions obtained by using the model in Table 1 and the accurate full-wave simulations conducted with Ansys HFSS. We note, however, that the simplified model for the RIS in Table 1 predicts the direction of the reflected beam of interest well. More precisely, we observe that a hotspot can be clearly observed in correspondence to the location of the tag (circled in purple in Fig. 5), which confirms the signal enhancement provided by the RIS at the target location. Also, this is in agreement with the experimental results reported in Fig. 3. The mismatch between the simple model in Table 1 and the full-wave simulations in Ansys HFSS can be reduced by enriching the model for the RIS proposed in Table 1 at the expense of higher modeling and optimization complexity. Also, it needs to be mentioned that the tag in Fig. 5 is located near the boundary of Fraunhofer's far-field, which in part justifies the mismatch in Fig. 5.

CONCLUSION

In this article, we have introduced and characterized a prototype of an RIS whose unit cells can be digitally and individually tuned through a control voltage. By appropriately configuring the control voltage, the phase of the EM field scattered by each unit cell can be optimized. We have characterized the electromagnetic behavior of the individual unit cells of the RIS under the assumption of periodic boundary conditions, and have proposed a simple voltage-dependent model for the reflection coefficient of the unit cells. We have shown that if the angles of incidence and reflection are confined within the range 40° , the unit cells can be assumed to have an almost omnidirectional and unit-gain radiation pattern, while the amplitude of the reflection coefficient depends on the applied phase shift through the control voltage. We have compared the proposed model against full-wave simulations, and have shown that, despite simple, the model can be successfully employed for first-order optimization of an RIS. By capitalizing on the proposed model for the RIS and optimization method for computing a location-dependent RIS beamforming codebook, we have substantiated, through experimental measurements, that an RIS can enhance the performance and reliability of AmBC.

In this article, we have introduced and characterized a prototype of an RIS whose unit cells can be digitally and individually tuned through a control voltage. By appropriately configuring the control voltage, the phase of the EM field scattered by each unit cell can be optimized.

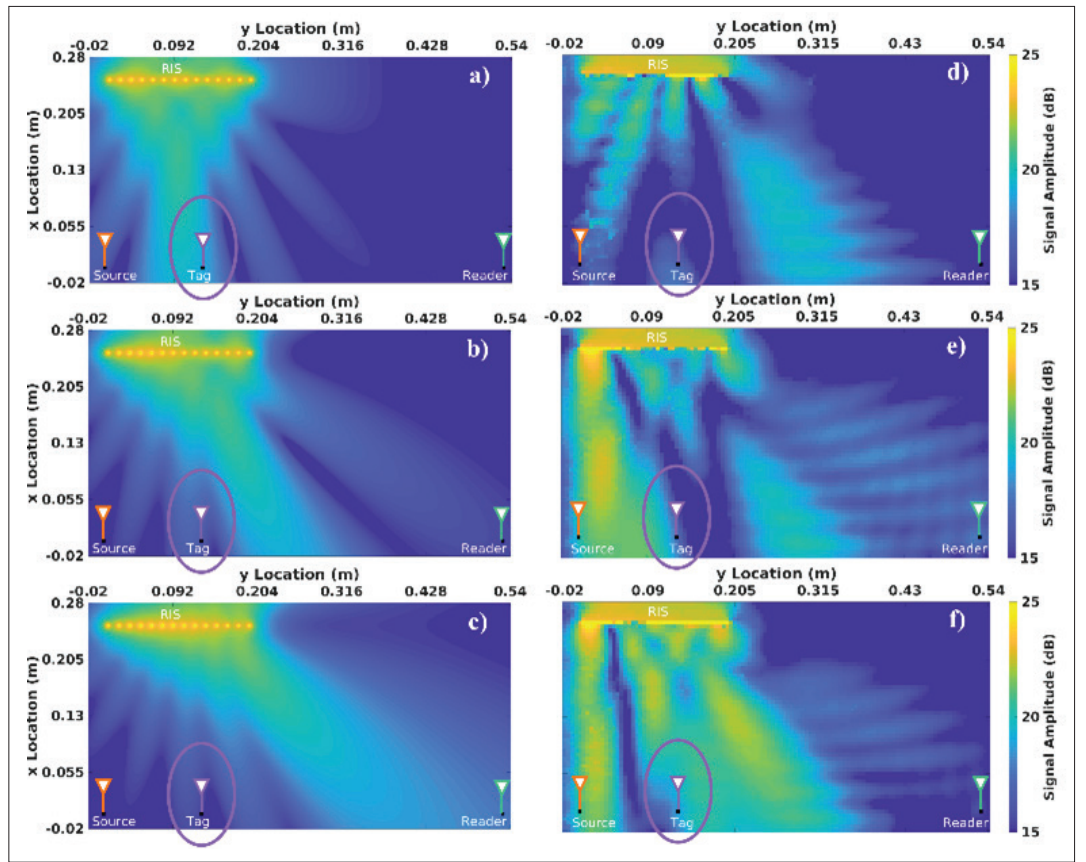


FIGURE 5. Illustration of the field scattered by the RIS for three different beamforming pairs $(b_m^{(P)}, \psi^{(P)})$ that minimize the BER in Fig. 3; a–c show the results obtained with the simple model in Table 1; d–f show the results obtained through full-wave simulations in Ansys HFSS.

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BIOGRAPHIES

ROMAIN FARA (romain.fara@centralesupelec.fr) received his Master's degree in electrical engineering from the National Institute of Applied Sciences of Lyon in 2018 and his Ph.D. degree from CentraleSupélec, Paris-Saclay University, France, in collaboration with Orange Innovation Networks, Chatillon, France, in 2021. His current research interests include ambient backscatter communications.

PHILIPPE RATAJCZAK (philippe.ratajczak@orange.com) received his M.Sc. and Ph.D. degrees in electrical engineering from the University of Nice-Sophia Antipolis in 1990 and 1995, respectively. In 1995, he joined CNET (now Orange Innovation Networks), Sophia-Antipolis, France. Since 2014, he has been the co-head of CREMANT, the joint research center on antennas between Orange Labs, the University of Nice, and CNRS. His research interests include reflectarrays, reconfigurable antennas with beamforming capabilities, and electromagnetic modeling and simulation.

DINH-THUY PHAN-HUY (dinhthuy.phanhuy@orange.com) received her degree in engineering from Supelec in 2001 and her Ph.D. degree in electronics and telecommunications from the National Institute of Applied Sciences (INSA) of Rennes, France, in 2015. In 2001, she joined France Telecom R&D (now Orange Innovation), France, where she currently works as a research project manager. She is currently participating in two European projects on 6G. Her research interests include wireless communications and beamforming.

ABDELWAHEB OURIR (a.ourir@espci.fr) received his degree in engineering from ENIT, Tunis, Tunisia, in 2003, and his Ph.D. degree in physics from Paris-Sud University, Orsay, France, in 2007. Since 2008, he has been a CNRS research engineer with the Institut Langevin, Paris, France. His current research interests include wireless communications, electromagnetic subwave-length imaging, and wave propagation in artificial materials.

MARCO DI RENZO [F] (marco.di-renzo@universite-paris-saclay.fr) received his Ph.D. degree in electrical engineering from the University of L'Aquila, Italy, in 2007. He is a CNRS research director with the Laboratory of Signals and Systems at Paris-Saclay University, France. He serves as the Editor-in-Chief of *IEEE Communications Letters*. He is a Fellow of IET and AAIA, an ordinary member of the European Academy of Sciences and Arts and the Academia Europaea, a Fulbright Fellow, and a Highly Cited Researcher.

JULIEN DE ROSNY (julien.derosny@espci.fr) received his M.S. and Ph.D. degrees in wave physics from UPMC, Paris, France, in 1996 and 2000, respectively. From 2000 to 2001, he was a postdoctoral fellow with ScriUSA. In 2001, he joined the CNRS at Laboratoire Ondes et Acoustique, France. Since 2014, he has been a CNRS research director with the Institut Langevin, Paris. His research interests include communications in complex media, and acoustic and electromagnetic waves-based imaging.